

Automated BGP/OSPF Network Deployment with Terraform and Ansible

Project Evidence Report

TELE 6420 — Final Project • Northeastern University

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This report presents evidence of a fully deployed BGP/OSPF network on AWS using Terraform for infrastructure provisioning and Ansible for router configuration. Three EC2 instances running FRRouting (FRR) 8.5.2 simulate real routers configured with OSPF (Area 0) within AS 100 and eBGP peering between AS 100 and AS 200.

The following sections include screenshots demonstrating: Terraform infrastructure outputs, Ansible playbook execution, OSPF neighbor relationships, BGP session establishment, routing table verification, and failure simulation with reconvergence.

1. Terraform Infrastructure — AWS Outputs

Terraform provisioned the VPC (10.0.0.0/16), three subnets, an Internet Gateway, security groups, and three EC2 instances. The outputs below confirm all instance IPs and subnet IDs were correctly assigned.

```
# module.vpc.aws_vpc.main:
resource "aws_vpc" "main" {
  arn                                = "arn:aws:ec2:us-east-1:122610482256:vpc/vpc-0e9ea5f52b7be9b24"
  assign_generated_ipv6_cidr_block   = false
  cidr_block                         = "10.0.0.0/16"
  default_network_acl_id            = "acl-0343e0ee481226ac53"
  default_route_table_id            = "rtb-08f3f58cf9252a505"
  default_security_group_id         = "sg-0fb4afe32536ea844"
  dhcp_options_id                   = "dopt-0a2205f3ce1326b9d"
  enable_dns_hostnames               = true
  enable_dns_support                 = true
  enable_network_address_usage_metrics = false
  id                                = "vpc-0e9ea5f52b7be9b24"
  instance_tenancy                   = "default"
  ipv6_netmask_length                = 0
  main_route_table_id               = "rtb-08f3f58cf9252a505"
  owner_id                           = "122610482256"
  tags                               = {
    "Name" = "frr-vpc"
  }
  tags_all                           = {
    "Name" = "frr-vpc"
  }
}

Outputs:
r1_private_ip = "10.0.1.8"
r1_public_ip  = "190.31.196.80"
r2_private_ip = "10.0.1.13"
r2_public_ip  = "3.237.239.253"
r3_private_ip = "10.0.2.5"
r3_public_ip  = "3.231.24.87"
subnet_a_id   = "subnet-02133d45ed7387c01"
subnet_b_id   = "subnet-0db01be837c52f92b"
subnet_c_id   = "subnet-0696b95954665f476"
vpc_id        = "vpc-0e9ea5f52b7be9b24"
(base) chetan@MacBook-Pro terraform %
```

Figure 1: terraform show output — VPC, subnet IDs, and all router public/private IPs

2. Ansible Playbook Execution

The Ansible playbook (site.yml) applies three roles: base (FRR installation), ospf (OSPF configuration on R1 and R2), and bgp (BGP configuration on R1 and R3). Both runs below show zero failures across all routers.

```
((base) chetan@MacBook-Pro ansible % ansible-playbook -i inventory/hosts.ini site.yml

PLAY [Base configuration for all routers] *****

TASK [Gathering Facts] *****
ok: [R1]
ok: [R3]
ok: [R2]

TASK [base : Enable IP forwarding] *****
ok: [R1]
ok: [R3]
ok: [R2]

TASK [base : Persist IP forwarding in sysctl.conf] *****
ok: [R1]
ok: [R2]
ok: [R3]

TASK [base : Remove all FRR repo files] *****
changed: [R2]
changed: [R1]
changed: [R3]

TASK [base : Remove cached RPMs to force fresh download] *****
changed: [R1]
changed: [R2]
changed: [R3]

TASK [base : Download libyang2 RPM (required for FRR 8.5)] *****
changed: [R1]
changed: [R3]
changed: [R2]

TASK [base : Download FRR 8.5.2 RPM] *****
changed: [R1]
changed: [R3]
changed: [R2]

TASK [base : Download FRR 8.5.2 RPM] *****
changed: [R1]
changed: [R3]
changed: [R2]
```

Figure 2: Ansible playbook run — base role tasks: IP forwarding, FRR 8.5.2 installation on all routers

Figure 3: Ansible playbook run — OSPF and BGP roles applied; PLAY RECAP shows 0 failed on R1, R2, R3

OSPF Area 0 is configured between R1 (10.0.1.8) and R2 (10.0.1.13) on Subnet A (10.0.1.0/28). Both routers reached Full adjacency. R1 is elected as Designated Router (DR) and R2 as Backup DR (BDR). The non-broadcast network type with explicit unicast neighbors was required due to AWS VPC's lack of multicast support.

Figure 4: show ip ospf neighbor detail on R1 — neighbor 10.0.1.13 (R2) in Full/DR state, 34m uptime, 7 state changes

eBGP peering is established between R1 (AS 100, 10.0.1.8) and R3 (AS 200, 10.0.2.5). R1 advertises the 10.0.1.0/28 prefix into BGP. R3 advertises 10.0.3.0/24. The BGP table on both routers confirms route exchange.

```
(base) chetan@MacBook-Pro bgp-ospf-project % ssh -i ~/.ssh/bgp-ospf-key.pem ec2-user@100.31.196.80 "sudo vtysh -c 'show ip bgp'"
ssh -i ~/.ssh/bgp-ospf-key.pem ec2-user@100.31.24.87 "sudo vtysh -c 'show ip bgp'"
BGP table version is 1, local router ID is 10.0.1.8, vrf id 0
Default local pref 100, local AS 100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: 0NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

  Network          Next Hop           Metric LocPrf Weight Path
  *> 10.0.1.0/28    0.0.0.0                0         32768 i

Displayed 1 routes and 1 total paths
BGP table version is 1, local router ID is 10.0.2.5, vrf id 0
Default local pref 100, local AS 200
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: 0NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

  Network          Next Hop           Metric LocPrf Weight Path
  *> 10.0.1.0/28    10.0.1.8                0         0 100 i
  10.0.3.0/24      0.0.0.0                0         32768 i

Displayed 2 routes and 2 total paths
```

Figure 5: show ip bgp on R1 and R3 — R1 advertises 10.0.1.0/28; R3 has learned 10.0.1.0/28 via 10.0.1.8 (AS path: 100)

5. Network Verification Script — All Routers

The automated verification script (scripts/verify_network.py) SSHes into all three routers and checks OSPF neighbor state, BGP session status, routing tables, and ping reachability. All checks passed on all routers.

```
((base) chetan@MacBook-Pro bgp-ospf-project % python3 scripts/verify_network.py

● BGP/OSPF Network Verification Script
=====

=====
R1 (100.31.196.80)
=====

[INFO] OSPF Neighbors
Neighbor ID Pri State Up Time Dead Time Address Interface RXmtl RqstL DBaml
10.0.1.13 1 Full/DR 26.487s 38.749s 10.0.1.13 eth0:10.0.1.8 0 0 0
[PASS] OSPF neighbor(s) in Full state

[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.1.8, local AS number 100 vrf-id 0
BGP table version 1
RIB entries 1, using 192 bytes of memory
Peers 1, using 725 KiB of memory

Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd PfxSnt Desc
10.0.2.5 4 200 5 5 0 0 0 00:01:05 0 1 R3-eBGP-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, A - Babel, F - PBR, f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

K>* 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:01:12
O 10.0.1.0/28 [110/10] is directly connected, eth0, weight 1, 00:01:12
C>* 10.0.1.0/28 is directly connected, eth0, 00:01:12
K>* 169.254.169.254/32 [0/0] is directly connected, eth0, 00:01:11
[PASS] Ping to 10.0.1.13 successful
[PASS] Ping to 10.0.2.5 successful
```

Figure 6: Verification script — R1: OSPF Full/DR [PASS], BGP Established [PASS], Ping to R2 and R3 [PASS]

```
(base) chetan@MacBook-Pro bgp-ospf-project % cd ~/Desktop/bgp-ospf-project
python3 scripts/verify_network.py

● BGP/OSPF Network Verification Script
=====

=====
R1 (100.31.196.80)
=====

[INFO] OSPF Neighbors
Neighbor ID      Pri State           Up Time      Dead Time Address      Interface      RXmtL RqstL DBsmL
10.0.1.13        1 Full/DR         32m06s      38.683s 10.0.1.13      eth0:10.0.1.8      0      0      0
[PASS] OSPF neighbor(s) in Full state

[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.1.8, local AS number 100 vrf-id 0
BGP table version 1
RIB entries 1, using 192 bytes of memory
Peers 1, using 725 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ OutQ  Up/Down  State/PfxRcd  PfxSnt Desc
10.0.2.5      4      200     34      34       0      0  00:30:48    0/0          0      1 R3-eBGP-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, A - Babel, F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

K>* 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:32:49
O  10.0.1.0/28 [110/10] is directly connected, eth0, weight 1, 00:32:49
C>* 10.0.1.0/28 is directly connected, eth0, 00:32:49
K>* 169.254.169.254/32 [0/0] is directly connected, eth0, 00:32:49
[PASS] Ping to 10.0.1.13 successful
[PASS] Ping to 10.0.2.5 successful
```

Figure 7: R1 full verification output — OSPF neighbor 10.0.1.13 Full/DR (32m uptime), BGP with R3 Established (30m uptime)

```
=====
R2 (3.237.239.253)
=====

[INFO] OSPF Neighbors
Neighbor ID      Pri State           Up Time      Dead Time Address      Interface      RXmtL RqstL DBsmL
10.0.1.8         1 Full/Backup     32m11s      35.604s 10.0.1.8      eth0:10.0.1.13      0      0      0
[PASS] OSPF neighbor(s) in Full state

[INFO] BGP Summary
(no output)

[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, A - Babel, F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

K>* 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:33:08
O  10.0.1.0/28 [110/10] is directly connected, eth0, weight 1, 00:33:07
C>* 10.0.1.0/28 is directly connected, eth0, 00:33:08
K>* 169.254.169.254/32 [0/0] is directly connected, eth0, 00:33:08
[PASS] Ping to 10.0.1.8 successful
```

Figure 8: R2 verification — OSPF Full/Backup with R1 (32m uptime), routing table shows OSPF route O 10.0.1.0/28

```

=====
R3 (3.231.24.87)
=====

[INFO] OSPF Neighbors
(no output)

[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.2.5, local AS number 200 vrf-id 0
BGP table version 1
RIB entries 3, using 576 bytes of memory
Peers 1, using 725 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt Desc
10.0.1.8      4      100      34       35       0     0     0 00:30:57      1          1 R1-eBGP-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, A - Babel, F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

K>* 0.0.0.0/0 [0/0] via 10.0.2.1, eth0, 00:32:59
B  10.0.1.0/28 [20/0] via 10.0.1.8 inactive, weight 1, 00:30:56
S>* 10.0.1.0/32 [1/0] via 10.0.2.1, eth0, weight 1, 00:32:58
C>* 10.0.2.0/28 is directly connected, eth0, 00:32:59
K>* 169.254.169.254/32 [0/0] is directly connected, eth0, 00:32:59
[PASS] Ping to 10.0.1.8 successful

[INFO] Verification complete.

(base) chetan@MacBook-Pro bgp-ospf-project %

```

Figure 9: R3 verification — BGP Established with R1 (30m uptime), BGP route B 10.0.1.0/28 in routing table

6. Failure Simulation and Reconvergence

The failure simulation was performed by shutting down R1’s eth0 interface, which simultaneously brought down the OSPF link to R2 and the BGP session to R3. The three states (before, during, and after) are documented below.

6.1 BEFORE Failure — All protocols fully operational

```

((base) chetan@MacBook-Pro bgp-ospf-project % python3 scripts/verify_network.py
● BGP/OSPF Network Verification Script
=====

=====
R1 (100.31.196.80)
=====

[INFO] OSPF Neighbors
Neighbor ID  Pri State          Up Time      Dead Time Address      Interface      RXmtL  RqstL  DBmtL
10.0.1.13    1 Full/DR      26.487s      38.749s 10.0.1.13    eth0:10.0.1.8      0       0       0
[PASS] OSPF neighbor(s) in Full state

[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.1.8, local AS number 100 vrf-id 0
BGP table version 1
RIB entries 1, using 192 bytes of memory
Peers 1, using 725 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt Desc
10.0.2.5      4      200       5       5        0     0     0 00:01:05      0          1 R3-eBGP-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, A - Babel, F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

K>* 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:01:12
O  10.0.1.0/28 [119/10] is directly connected, eth0, weight 1, 00:01:12
C>* 10.0.1.0/28 is directly connected, eth0, 00:01:12
K>* 169.254.169.254/32 [0/0] is directly connected, eth0, 00:01:11
[PASS] Ping to 10.0.1.13 successful
[PASS] Ping to 10.0.2.5 successful

```

Figure 10 (BEFORE): R1 OSPF Full/DR with R2, BGP Established with R3, all pings successful

6.2 DURING Failure — R1 interface down

R1's eth0 was shut down via vtysh. R1 became unreachable over SSH. R2 lost its OSPF neighbor. R3's BGP session dropped to Connect state. The BGP route 10.0.1.0/28 was withdrawn from R3's routing table.

```

(base) chetan@MacBook-Pro: ~$ ssh -i ~/.ssh/bgp-ospf-key.pem ec2-user@100.31.196.88 "sudo vtysh -c 'show ip ospf neighbor' && sudo vtysh -c 'show ip route'"
ssh: connect to host 100.31.196.88 port 22: Operation timed out
(base) chetan@MacBook-Pro: ~$ ssh -i ~/.ssh/bgp-ospf-key.pem ec2-user@100.31.196.88 "cd ~/Desktop/bgp-ospf-project; python3 scripts/verify_network.py"
@ BGP/OSPF Network Verification Script
=====

R1 (100.31.196.88)
=====
[FAIL] Could not connect: timed out
=====

R2 (3.237.239.153)
=====
[INFO] OSPF Neighbors
Neighbor ID Pri State Up Time Dead Time Address Interface RxMtl Rgstr DBLen
[FAIL] No OSPF Full neighbors detected
[INFO] BGP Summary
(no output)
[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, A - Abbil, F - FIB, f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
K> 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:45:07
O> 10.0.1.0/28 [10/28] is directly connected, eth0, weight 1, 00:45:56
C> 10.0.1.0/28 is directly connected, eth0, 00:45:57
K> 169.254.169.254/32 [0/0] is directly connected, eth0, 00:45:57
[FAIL] Ping to 10.0.1.8 FAILED
=====

R3 (3.231.24.87)
=====
[INFO] OSPF Neighbors
(no output)
[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.2.5, local AS number 200 vrf-id 0
BGP table version 2
RIB entries 1, using 392 bytes of memory
Peers 1, using 725 KiB of memory
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd PfxSnt Desc
10.0.1.8 4 100 39 44 0 0 0 00:05:48 Connect 0 R1-BGP-peer
Total number of neighbors 1
[FAIL] No BGP Established sessions found
[INFO] Routing Table
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, A - Abbil, F - FIB, f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
K> 0.0.0.0/0 [0/0] via 10.0.2.1, eth0, 00:45:58
S> 10.0.1.0/28 [10] via 10.0.2.1, eth0, weight 1, 00:45:49
C> 10.0.2.0/28 is directly connected, eth0, 00:45:58
K> 169.254.169.254/32 [0/0] is directly connected, eth0, 00:45:58
[FAIL] Ping to 10.0.1.8 FAILED
=====
Verification complete.

```

Figure 11 (DURING): R1 SSH timed out; R2 shows no OSPF neighbors [FAIL]; R3 BGP dropped to Connect state [FAIL] — route 10.0.1.0/28 absent from R3 routing table

6.3 AFTER Reconvergence — R1 interface restored

R1 was rebooted via the AWS console to restore eth0. Within ~35 seconds, OSPF re-established Full adjacency between R1 and R2, and the BGP session between R1 and R3 came back up. The BGP route 10.0.1.0/28 reappeared in R3's routing table, confirming full reconvergence.


```

Peer 1, using 725 KiB of memory
Neighbor V AS MsgRcvd MsgSent TlvsRcvd TlvsSnd Up/Down State/PfxRcd PfxRcd Desc
10.0.2.5 A 200 0 0 0 00:01:00 0 1 R3-OSPF-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: C - learned route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
F - Table A - Mobile, F - PNM, F - OpenFabric,
* - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

K> 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:01:12
O 10.0.1.0/28 [10/0] is directly connected, eth0, weight 1, 00:01:12
C> 10.0.1.0/28 is directly connected, eth0, 00:01:12
K> 169.254.169.254/32 [0/0] is directly connected, eth0, 00:01:11
[PASS] Ping to 10.0.1.3 successful
[PASS] Ping to 10.0.2.5 successful

=====
R2 (10.207.209.200)
=====

[INFO] OSPF Neighbors
Neighbor ID Pri State Up Time Dead Time Address Interface RmtI RmtI DbmI
10.0.1.0 0 Full/Backup 36.823s 33.171s 10.0.1.0 eth0/10.0.1.13 0 0 0
[PASS] OSPF neighbor(s) in Full state

[INFO] BGP Summary
[No output]

[INFO] Routing Table
Codes: C - learned route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
F - Table A - Mobile, F - PNM, F - OpenFabric,
* - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

K> 0.0.0.0/0 [0/0] via 10.0.1.1, eth0, 00:09:58
O 10.0.1.0/28 [10/0] is directly connected, eth0, weight 1, 00:09:57
C> 10.0.1.0/28 is directly connected, eth0, 00:09:58
K> 169.254.169.254/32 [0/0] is directly connected, eth0, 00:09:58
[PASS] Ping to 10.0.1.8 successful

=====
R3 (10.221.24.87)
=====

[INFO] OSPF Neighbors
[No output]

[INFO] BGP Summary
IPv4 Unicast Summary (VRF default):
BGP router identifier 10.0.1.0, local AS number 200 vrf-id 0
BGP table version 3
BGP entries 0, using 526 bytes of memory
Peer 1, using 725 KiB of memory

Neighbor V AS MsgRcvd MsgSent TlvsRcvd TlvsSnd Up/Down State/PfxRcd PfxRcd Desc
10.0.1.0 A 200 40 0 0 00:01:10 1 1 R3-OSPF-peer

Total number of neighbors 1
[PASS] BGP session(s) Established

[INFO] Routing Table
Codes: C - learned route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
F - Table A - Mobile, F - PNM, F - OpenFabric,
* - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

K> 0.0.0.0/0 [0/0] via 10.0.2.1, eth0, 00:09:48
O 10.0.1.0/28 [10/0] via 10.0.1.8 inactive, weight 1, 00:01:13
O 10.0.1.0/28 [10/0] via 10.0.2.1, eth0, weight 1, 00:09:47
C> 10.0.2.0/28 is directly connected, eth0, 00:09:48
K> 169.254.169.254/32 [0/0] is directly connected, eth0, 00:09:48
[PASS] Ping to 10.0.1.8 successful

Verification complete.
(base) chetan@macbook-Pro: ~ % bgp-ospf-project %

```

Figure 12 (AFTER): All checks PASS — R1 OSPF Full/DR restored (36s uptime), R2 Full/Backup restored, R3 BGP re-Established (1m14s uptime), route B 10.0.1.0/28 back in R3 routing table

Reconvergence Evidence Summary

- OSPF reconverged: R2's OSPF neighbor table went from empty (during) to Full/Backup with R1 (after) within the dead timer interval (40s).
- BGP reconverged: R3's BGP session went from Connect (during) to Established (after), and the withdrawn route B 10.0.1.0/28 was re-advertised and re-installed in R3's routing table.
- Total reconvergence time: approximately 35–40 seconds after R1 was restored, consistent with OSPF dead timer and BGP hold timer behavior.